

DATA SCIENCE COURSE

Why Python for Data Science?

Lecturer SO Vengleab · April 2026

Introduction to Python

Python is a Programming language.

- You write **source code** — human-readable instructions.
- It is translated into **computer code** the machine executes.
- Together they become **software**.



THE BIG QUESTION

Is coding a must for Data Science?

Let's settle this before we write a single line.

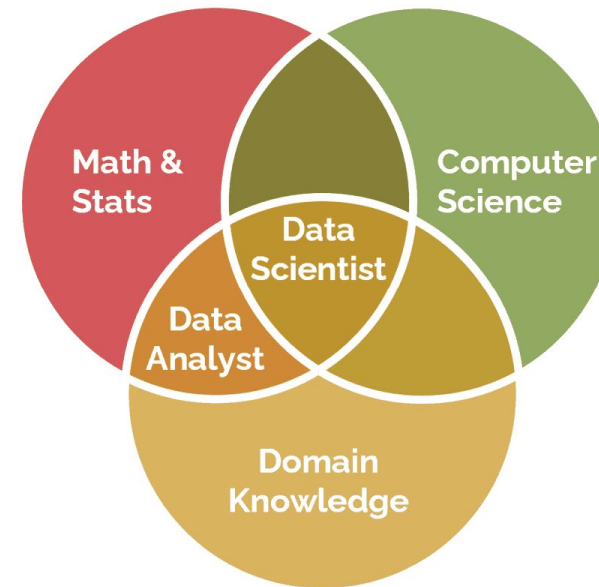
THE BIG QUESTION

The Answer is NO!

- You can do data science without writing code.
- No-code tools cover the basics: Excel, Google Sheets, Power BI, Tableau, KNIME.
- Drag-and-drop workflows can collect, clean, and chart data.



Venn Diagram of Data Science Field



● **Insight:** No-code gets you started fast — the question is what happens when the data grows and the task repeats.

What Data Scientists Do

01

Data Collection

Pull data from files, databases, APIs, and the web.

02

Preparation & Cleansing

Fix missing values, duplicates, and messy formats.

03

EDA

Explore, summarize, and visualize to find patterns.

04

Model Building

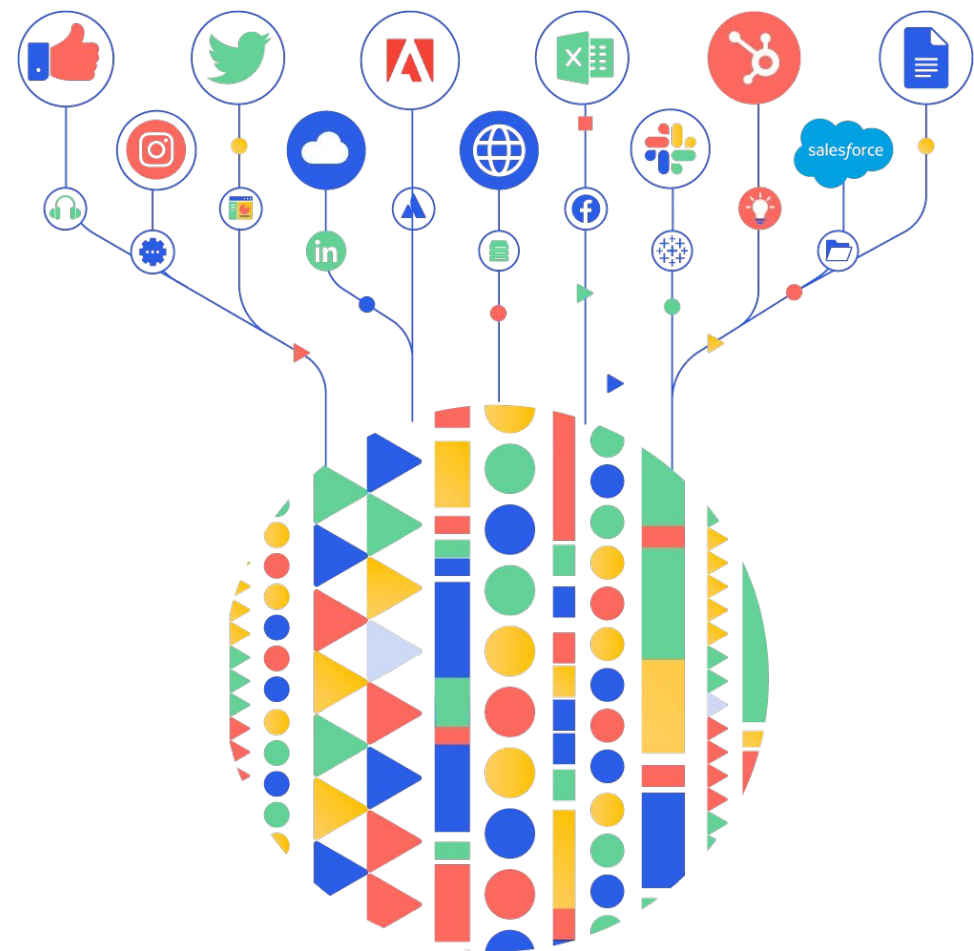
Train machine learning models — if the problem needs one.

05

Communicate

Share findings and collaborate with the team.

Data Collection



Data Preparation and Cleansing

#	Id	Name	Birthday	Gender	IsTeacher?	#Students	Country	City
1	111	John	31/12/1990	M	0	0	Ireland	Dublin
2	222	Mery	15/10/1978	F	1	15	Iceland	
3	333	Alice	19/04/2000	F	0	0	Spain	Madrid
4	444	Mark	01/11/1997	M	0	0	France	Paris
5	555	Alex	15/03/2000	A	1	23	Germany	Berlin
6	555	Peter	1983-12-01	M	1	10	Italy	Rome
7	777	Calvin	05/05/1995	M	0	0	Italy	Italy
8	888	Roxane	03/08/1948	F	0	0	Portugal	Lisbon
9	999	Anne	05/09/1992	F	0	5	Switzerland	Geneva
10	101010	Paul	14/11/1992	M	1	26	Ytali	Rome

Missing values

Invalid values

Misfielded values

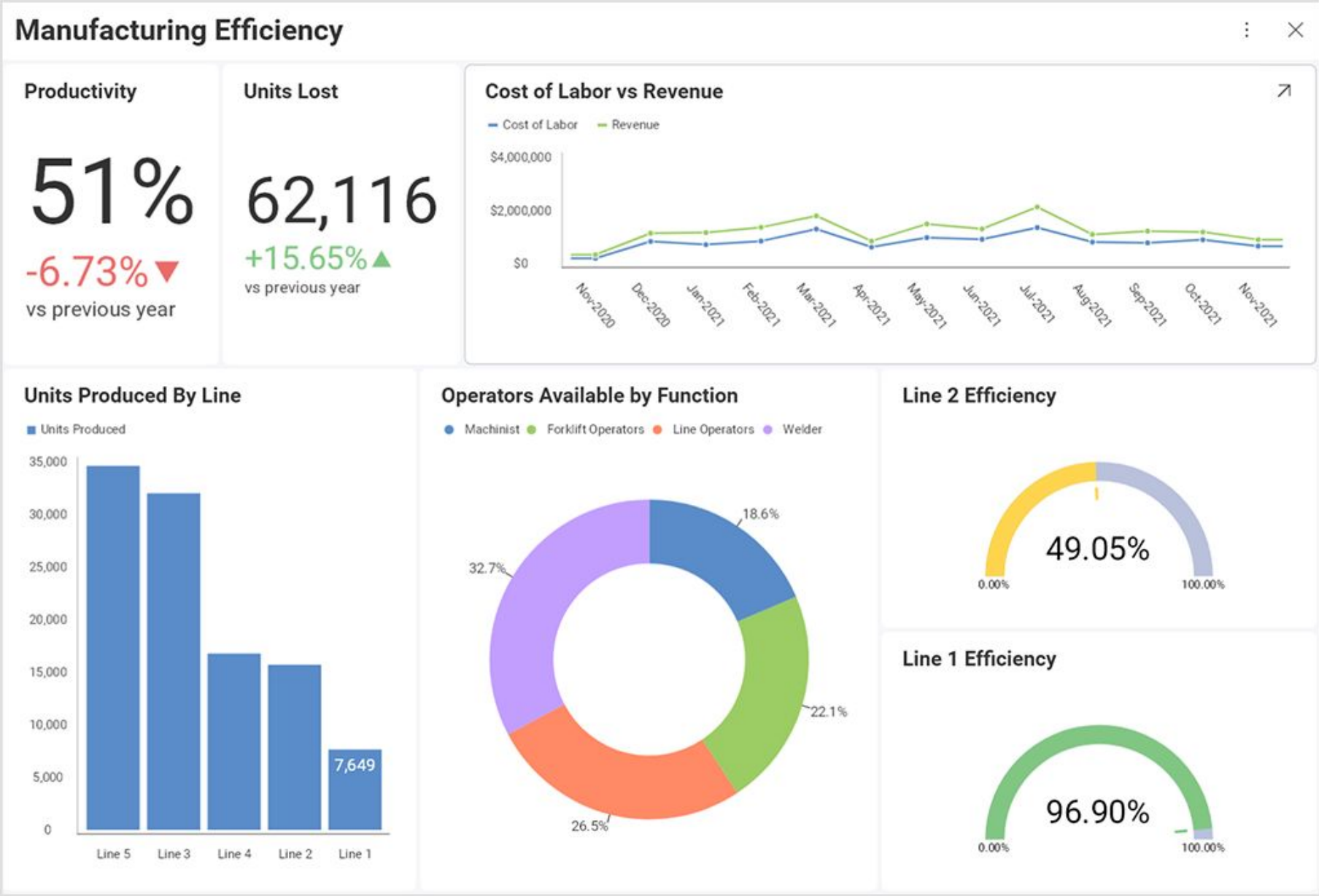
Misspellings

Uniqueness

Formats

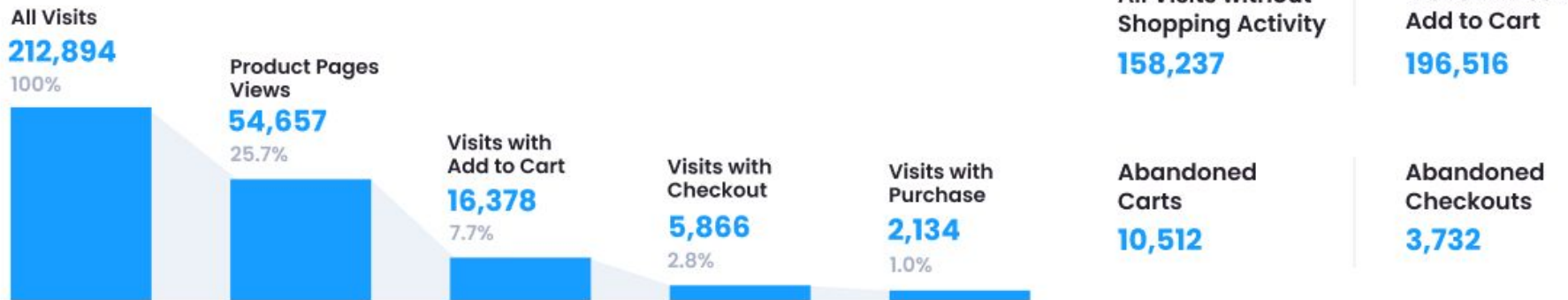
Attribute dependencies

Exploratory Data Analysis (EDA)

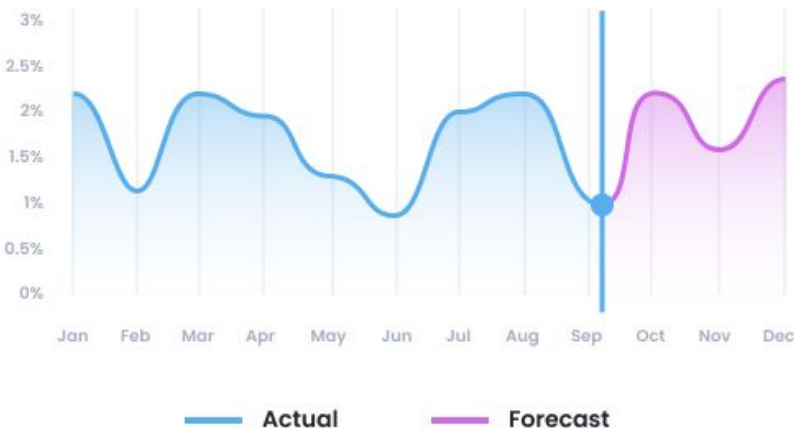


Exploratory Data Analysis (EDA)

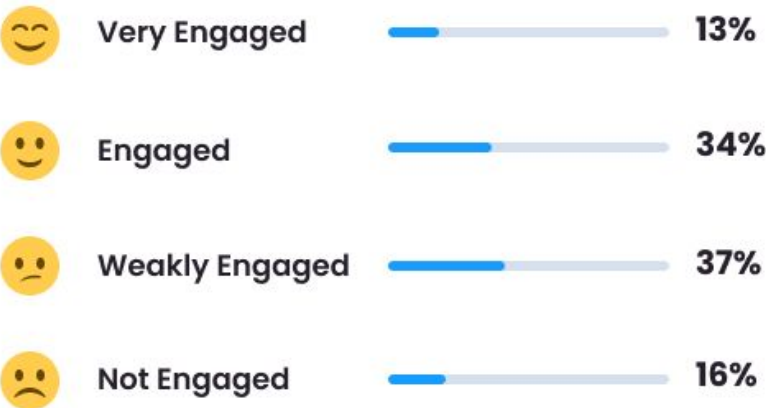
Shopping Behavior Analysis



Visitors to Customers Conversion Trend



Customer Sentiment Breakdown



Without Coding vs With Python

Without coding

- Quick start — click, drag, drop
- Limited to what the tool offers
- Hard to automate repeated work
- Struggles with large or messy data
- Workflows are hard to share and reproduce

With coding — especially Python

- One script handles collection to communication
- Automate anything, schedule it, rerun it
- Scales to millions of rows
- Full control over cleaning, EDA, and models
- Code is shareable, versioned, reproducible

GETTING STARTED

Why Python?

A one-slide definition — then your first program.

Why Python?

Simple, powerful, everywhere.

- **Open-source** and free, running on **Windows, Mac, and Linux**.
- **General purpose** — suitable for analysis of financial data and far beyond.
- **High-level**: human-readable code that beginners can pick up fast.



What is Python

Three things to know

- High-level language — Python reads almost like plain English, easy to learn and write.
- Created in 1991 — designed by Guido van Rossum; now one of the world's most popular languages.
- Used everywhere — web dev, data science, AI, automation, scripting.
- Your first Python program →



Insight: Simple. Powerful. Elegant — one line of Python is already a complete program.



```
# Hello, World!  
print("Hello, World!")  
  
# Output:  
# Hello, World!
```

Reference

- <https://www.python.org/about/>
- <https://docs.python.org/3/tutorial/>
- <https://pandas.pydata.org/>
- <https://scikit-learn.org/>